Marketing automation with AI & agents

Architecture governance | Working design for review | 25 October 2026

Business objectives and governing principles

AP-01 Risk-based oversight Increase automation without uncontrolled exposure. Require approval based on audience, discount exposure and customer risk.

AP-02 Privacy and least privilege Maintain customer trust and satisfy applicable PDPL obligations. Minimize model-visible data; enforce consent, purpose and access in trusted services.

AP-03 Reliable, auditable execution Prevent duplicate outreach and enable investigation. Use durable operation IDs, idempotent provider calls where available and reconciliation.

AP-04 Measurable, cost-aware AI Improve marketing relevance within budget. Route rules and models by measured quality, latency and cost.

Architecture decision records

ADR-001 | Hybrid campaign processing
Real-time only: fresh but expensive; batch only: inexpensive but too slow for triggers; hybrid: separate operating paths. **Decision:** nightly segmentation on isolated analytics data, event-driven cart triggers. **Cost:** two paths to monitor and operate.

ADR-002 | Tiered model routing
Managed API: low operations, variable fees; self-hosted: more control, GPU overhead; hybrid: flexible but complex. **Decision:** deterministic rules first, smaller models where benchmarks justify them, managed models for complex work subject to privacy review.

ADR-003 | Structured and semantic retrieval
Vector-first: unsuitable for exact order facts; structured-only: weak semantic matching; hybrid: synchronization overhead. **Decision:** authoritative customer and consent stores plus warehouse analytics; optional vector index for approved product/brand text after retrieval evaluation.

Architecture Review Board | Evidence required

Security	Service identity, least privilege, input validation, injection tests
Privacy / PDPL	Jurisdiction, lawful purpose, consent and opt-out, retention, residency
Evaluation	Content quality, retrieval relevance, policy-bypass regression tests
Cost / resilience	p95 latency, token spend, alerts, retry and reconciliation tests
Human oversight	Approval ownership, scope and version binding; revocation and audit

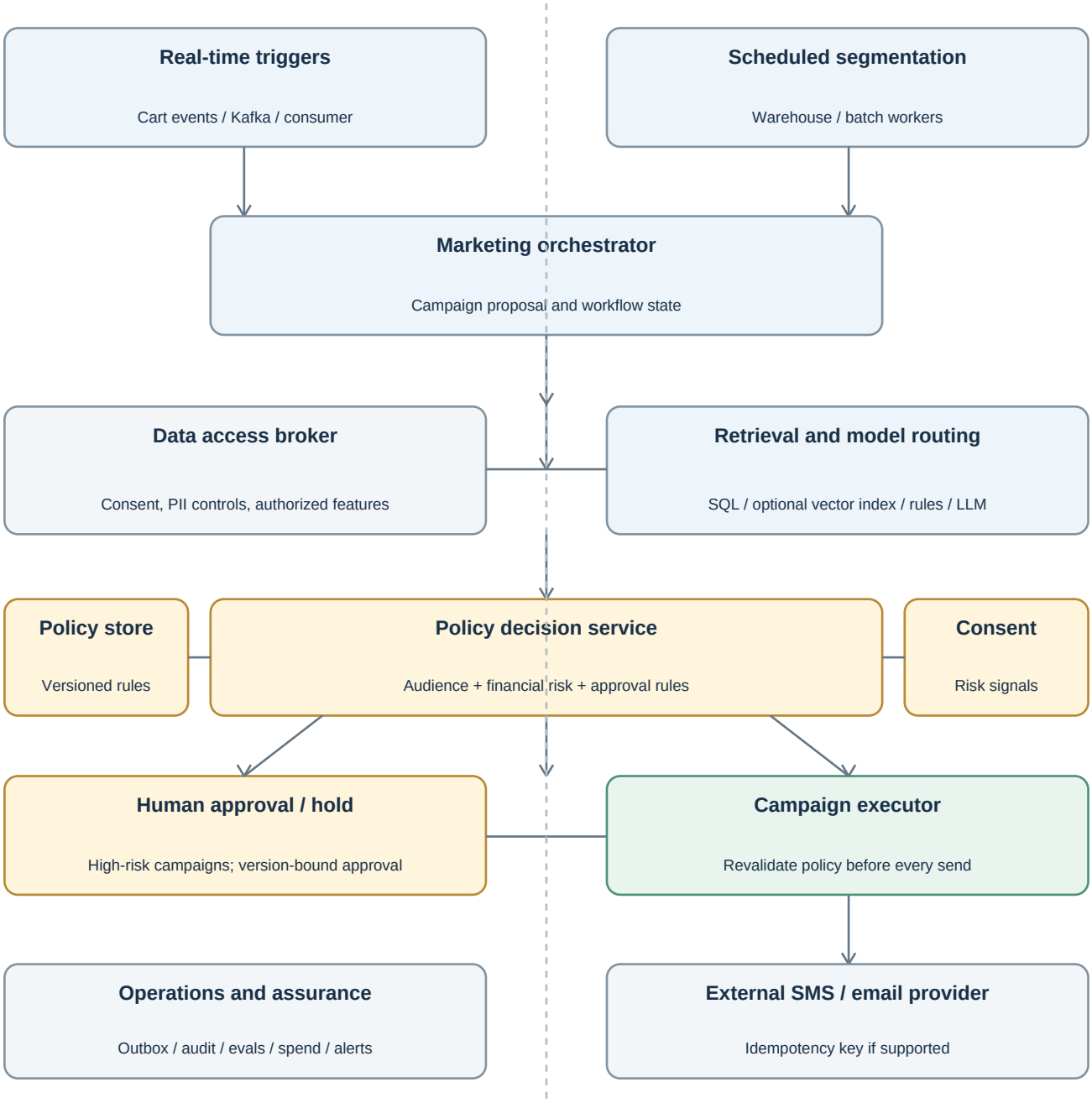
Automated policy guardrail

Changes to **campaign-policy.yaml** trigger CI tests for risk thresholds and high-exposure scenarios. Protected branches require two designated code-owner approvals; policy tests and deployment checks cannot be bypassed by ordinary contributors. Production policy changes are versioned, logged and reversible.

Open decisions: confirm PDPL jurisdiction and approval owners; benchmark model costs and latency; verify provider idempotency; define whether the five-second SLA covers submission or delivery.

Reference architecture and trust boundaries

Two processing paths; authorization remains outside the model.



Boundary: the LLM cannot resolve raw PII, approve campaigns or call providers directly.

If provider outcome is uncertain, reconcile before retrying to reduce duplicate sends.